

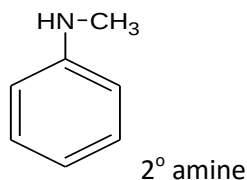
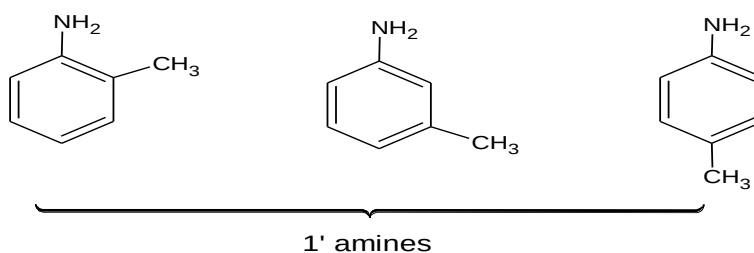
Chapter 11: Amines

PSPM 2011 / 2012

- Q.1 (a) Draw the structural formula of the aromatic amines with a molecular formula C_7H_9N . Classify them into primary and secondary amines. Suggest a chemical test to differentiate between a chosen primary and the secondary amines. State the observation and write the chemical equations involved.

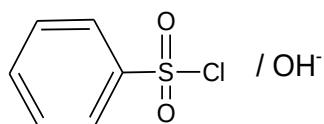
10 m

Answer (a)

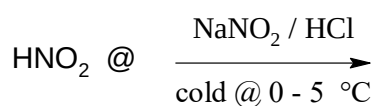


Hinsberg's test

(a)

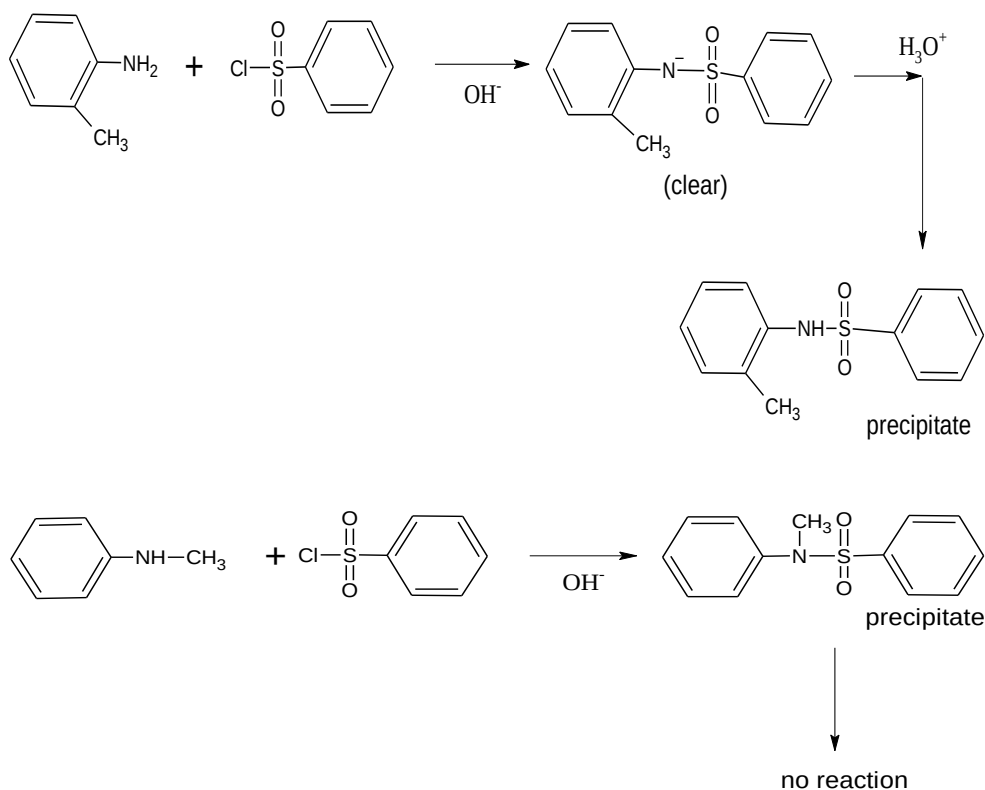


Or

(b) H_3O^+

* include equation & observation

Observation	Initial	Final
1° amine	Clear solution forms	Precipitate forms
	Precipitate forms	Precipitate remains

equation

PSPM 2012 / 2013

- Q. 2 (a) Arrange the following amines in ascending order of basicity. Explain your answer. 10 m

aniline, propylamine, N-methylethanamine, N,N-dimethylethanamine

Indicate an amine that can be prepared through Hoffmann's degradation. Write the chemical equation for this reaction.

Suggest a chemical test to differentiate between propylamine and *N*-methylethanamine. State the observation and write the chemical equation involved.

Answer (a) Aniline < *N,N*-dimethylethanamine < propylamine < *N*-methylethanamine

Note:

Aromatic vs aliphatic amines – 1M

Between aliphatic amines – 1M

Explanation

Aniline is least basic due to the lone pair on N **delocalise** into the benzene ring therefore less available to accept proton.

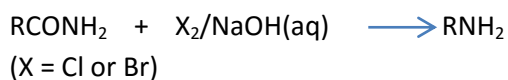
N-methylethanamine is most basic due to **inductive effect** OR the presence of methyl group increase availability of electron to accept proton

OR

N,N-dimethylethanamine is least basic due to **steric effect** OR difficult for N to accept proton.

Propylamine OR aniline

Hoffmann's degradation:



Note:

Starting material – 1M Reagent – 1M

Hinsberg's test OR nitrous acid

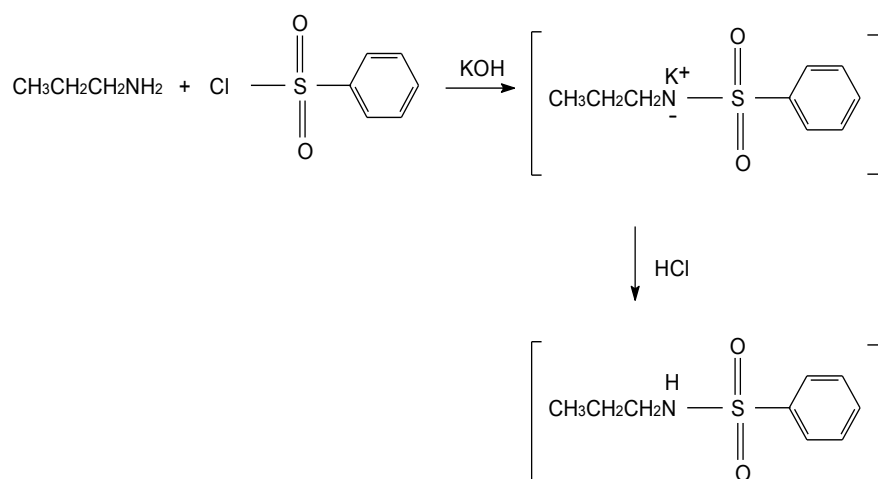
Observation: Hinsberg's Test

Propylamine – Clear solution, on acidification precipitate formed

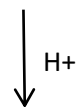
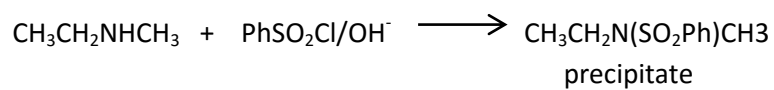
OR

N-methylethanamine – White precipitate, precipitate remain on acidification.

Equation:



OR



Precipitate remains

OR

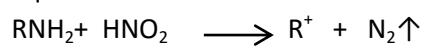
Observation : Nitrous acid

Propylamine – effervescence OR bubbles of gas

OR

N-methylethanamine – Yellow oil formed

Equation:



OR



PSPM 2013 / 2014

- Q. 3 (a) Draw the structure of a primary, secondary and tertiary amine with molecular formula $C_4H_{11}N$. using suitable amines, explain basicity in terms of inductive effect and resonance effect. 12 m
- Suggest an amine that can be prepared from a nitrile compound. Write the chemical equation for this reaction.

Answer (a) $1^\circ - CH_3CH_2CH_2CH_2NH_2$ or $(CH_3)_3CNH_2$

$2^\circ - CH_3CH_2CH_2NH(CH_3)$ or $CH_3CH_2NHCH_2CH_3$ or $CH_3NHCH(CH_3)_2$

$3^\circ - (CH_3)_2NHCH_2CH_3$

Inductive effect: 2° amine more basic than 1° amine

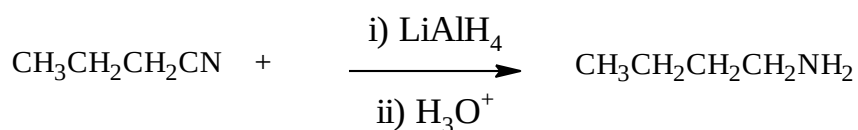
More alkyl groups (electron releasing groups) attached to the nitrogen of the 2° alkyl amine.

The electron density is greater around the nitrogen atom and can release electrons/accept protons easily in the 2° alkyl amine.

Resonance effect: aniline (aromatic amine) less basic than the primary, secondary and tertiary aliphatic amine.

Aniline is strongly stabilized by resonance OR lone pair electrons on N atom are delocalised into the benzene ring.

1° amine



Note: The following reagents can also be used: $NaBH_4$ in methanol or H_2 /catalyst

PSPM 2014 / 2015

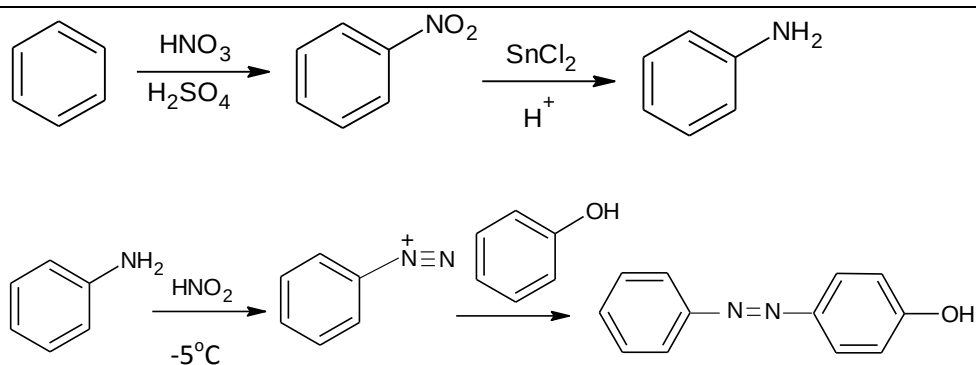
- Q. 4 (a) Aniline is one of the important starting material in chemical industry. Starting from benzene, show how aniline can be prepared.

13 m

One of the industrial applications of aniline is in the production of azo compound. Write the synthetic pathway for the formation of azo compound from aniline.

Aniline and cyclohexanamine have pK_b values of 9.4 and 3.3, respectively. Based on these values, compare and explain the basicity of the amines.

Answer (a)

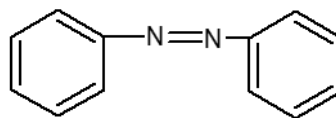


Cyclohexanamine is more basic @ aniline is less basic

Phenyl group is electron withdrawing group with the resonance effect which delocalised electron density at the nitrogen atom hence decreasing basicity.

PSPM 2015 / 2016

- Q. 5 (a) By using amines of your choice, explain the basicity of primary aliphatic amines, secondary aliphatic amines and aromatic amines. p-(Phenylazo)phenol, an orange dye and its structure is shown below : 10 m



p-(Phenylazo)phenol

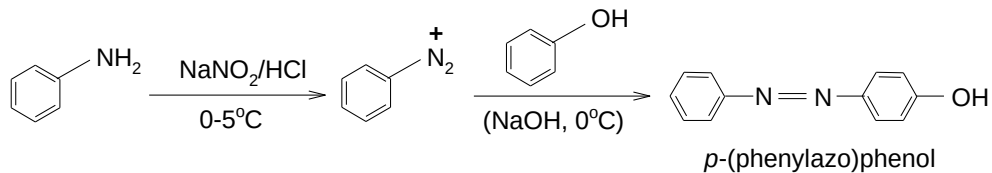
Show a synthetic pathway for the preparation of p-(phenylazo)phenol from aniline.

Answer (a) R_2NH more basic than RNH_2 .

Presence of electron donating substituent R, increase the inductive effect, increase basicity.

Aniline, aromatic amine less basic compared to aliphatic amine.

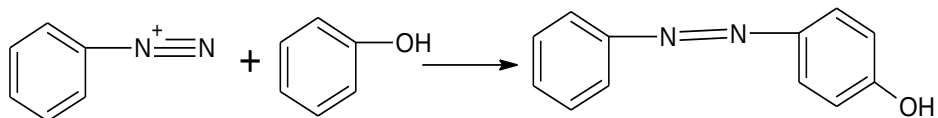
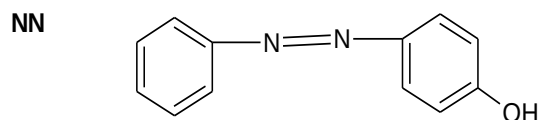
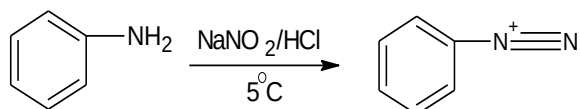
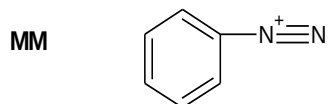
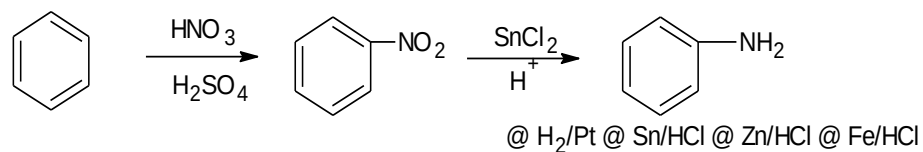
Electrons delocalized into the benzene ring of aniline, thus reducing the basicity.



PSPM 2016 / 2017

- Q. 6 (a) Show how would you prepare aniline from benzene. 9 m
- Aniline reacts with acidified sodium nitrate at 5°C to form **MM**. An addition of phenol to **MM** produces **NN**. Write the chemical equations involved and draw the structures of **MM** and **NN**. Provide a significant physical feature and a use of compound **NN**.

Answer (a)



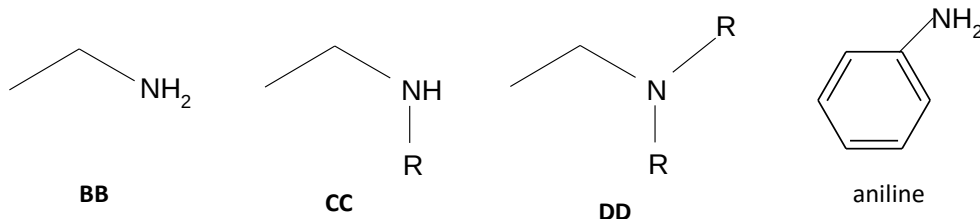
Significant feature: Intense / bright / orange colour

Application: Use as a dye @ as a colorant

PSPM 2017 / 2018

- Q. 7 (a) A group of students runs one simple experiment to distinguish three different classes of aliphatic amines and aniline. **BB**, **CC** and **DD** are primary, secondary and tertiary aliphatic amines, respectively. By using specific examples of **BB**, **CC**, and **DD**, suggest a chemical test to differentiate between:
- (i) **BB**, **CC** and **DD**
- (ii) aliphatic amine and aniline
- Arrange **BB**, **CC** and aniline in order of increasing basicity. Explain your answer.

Answer (a)



Hinsberg test:

1°: addition of benzenesulphonyl chloride in excess KOH, clear solution formed. Addition of HCl, white precipitate formed.

2°: addition of benzenesulphonyl chloride in excess KOH, white precipitate formed. After addition of HCl, white precipitate remained / no observable change.

3°: Clear solution formed when benzenesulphonyl chloride and excess HCl was added.

@ Nitrous acid:

1°: bubble gas

2°: yellow oil

3°: clear solution

(i) Bromine water:

Aliphatic amines: no white precipitate formed.

Aniline: white precipitate formed.

@ Nitrous acid:

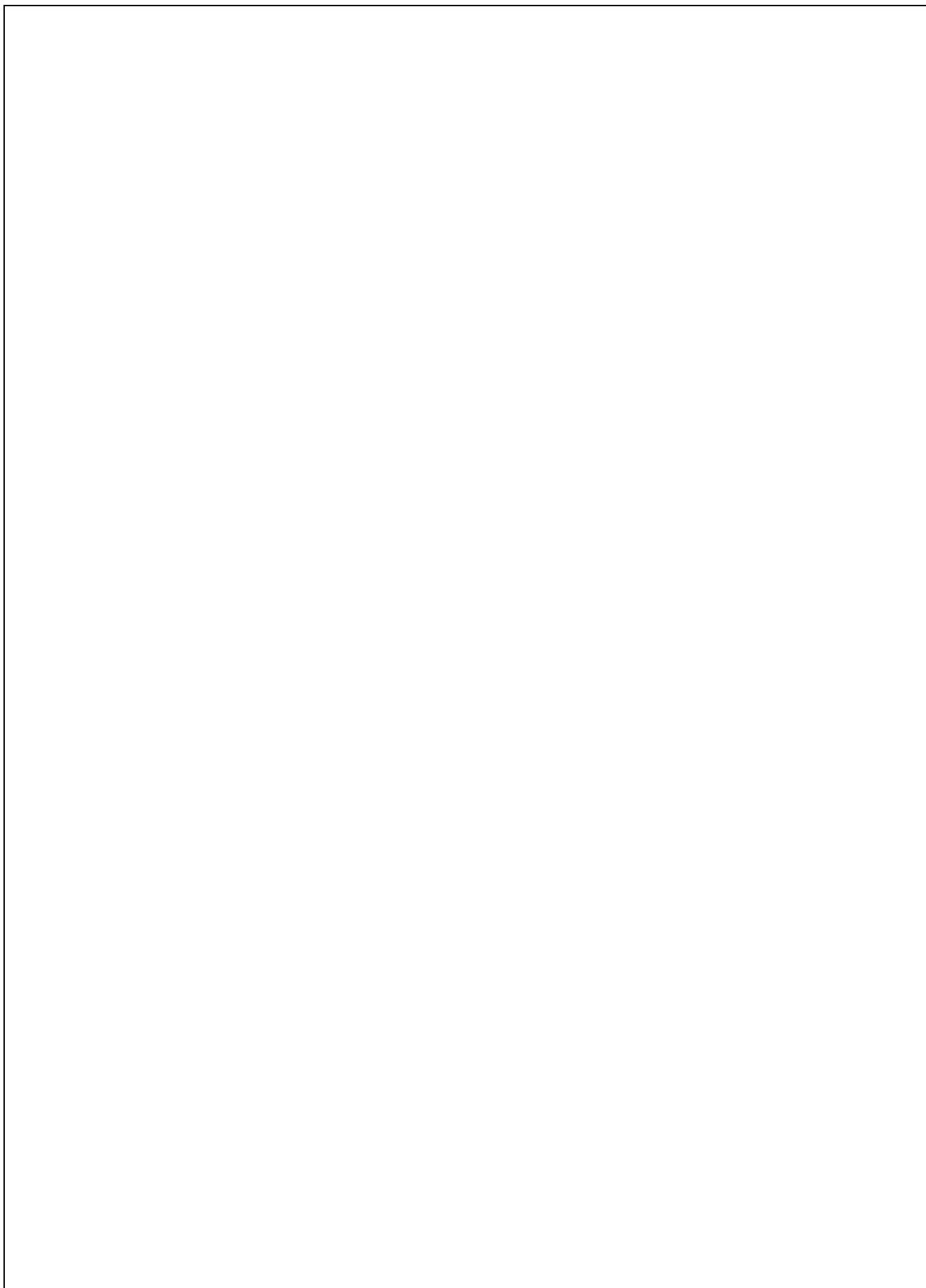
Aliphatic amines: 1°: bubble gas, 2°: yellow oil

Aniline: clear solution / no observable change / no gas bubble formed.

Increasing basicity: aniline, BB, CC

Alkyl group BB and CC is electron donating group which stabilize the positive charge @ increase the electron density on nitrogen.

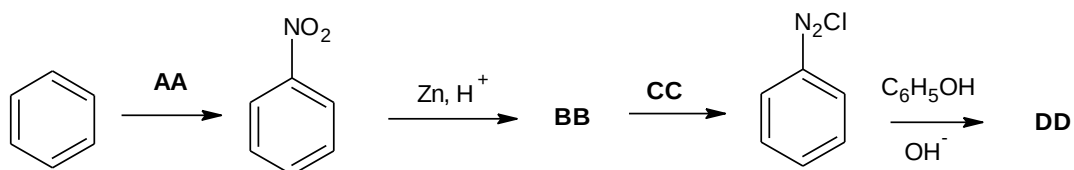
Aniline is weaker base due to delocalization of electron in the benzene ring destabilize the positive charge on nitrogen @ reduce electron density on nitrogen.



PSPM 2018 / 2019

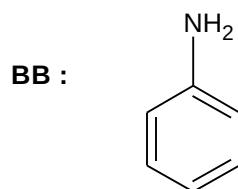
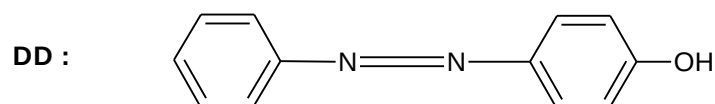
Q. 8 (a) The reaction scheme below shows the formation of a dye DD from benzene.

8 m



(i) Give the reagents AA and CC and draw the structural formulae of BB and DD.

Answer

AA : conc. HNO_3 , conc. H_2SO_4 , $50-55^\circ\text{C}$ **CC** : NaNO_2 , HCl , $< 5^\circ\text{C}$ 

(ii) Name the reagent for the formation of DD.

Answer

Coupling reaction

(b) N-methylamphetamine is a potent central nervous system stimulant that is mainly used as a recreational drug. The structure of N-methylamphetamine is shown below.

Name a chemical test to distinguish N-methylamphetamine from aniline. State the observations and write the chemical equation.

Answer b.

Hinsberg test

Observations:

Aniline :
Addition of benzenesulphonylchloride

↓ excess NaOH
clear solution

↓ HCl
white precipitate

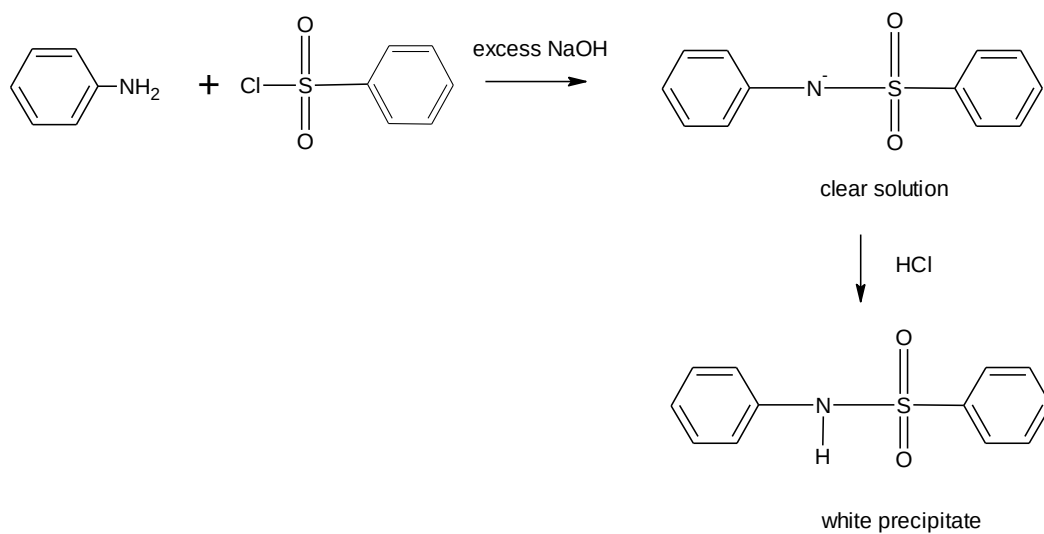
N - methylamphetamine
Addition of benzenesulphonylchloride

↓ excess NaOH
white precipitate

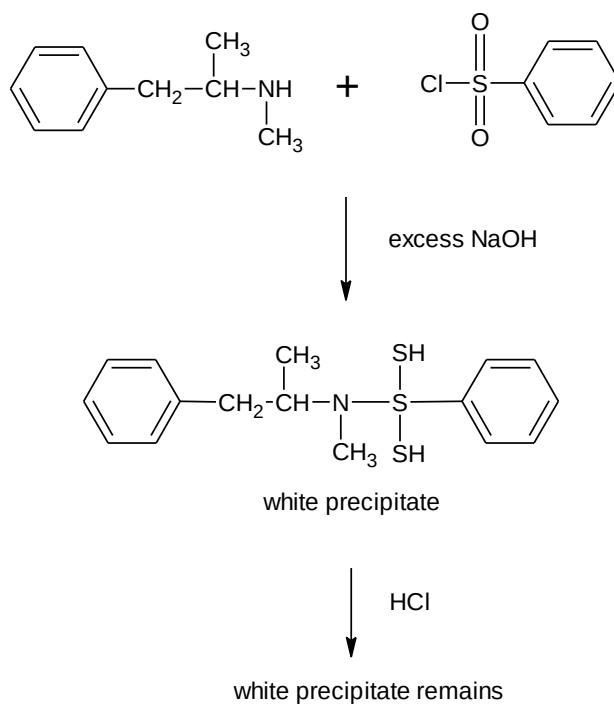
↓
white precipitate remains @ no observable change

Equation :

Aniline :



N-methylamphetamine :



OR

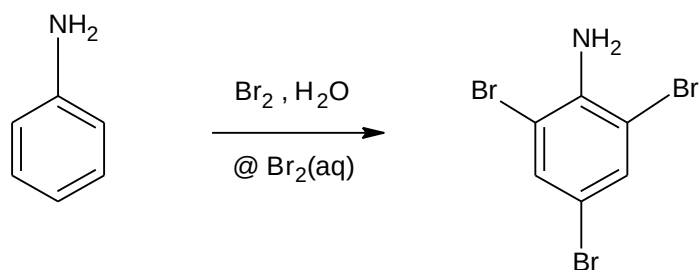
Bromine water

Observation :

Aniline : white precipitate

N-methylamphetamine : no observable change / no white precipitate form.

Equation:



OR

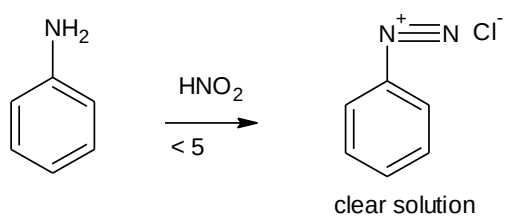
Nitrous acid

Observations :

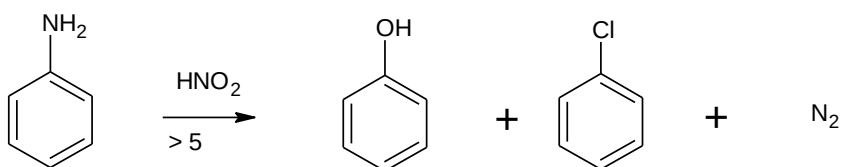
Aniline : Clear solution/no yellow oil formed/effervescence/bubble of gas above 5 °C.

N-methylamphetamine : Yellow oil formed.

Equation:



@



@

